RSTAT and RROLL Boundary: Analysis and Eligibility Universes

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Abstract

RSTAT and RROLL are reserved civic-evidence layers, not implemented packages. RSTAT is reserved for statistical and forensic analysis outputs: anomaly scores, uncertainty intervals, diagnostics, model transcripts, and caveats over normalized count, history, plan, and context inputs. RROLL is reserved for aggregate registration, eligibility, and ballot-style universes, with source hashes and privacy caveats. The boundary is intentionally conservative. RSTAT can identify patterns that need explanation. RROLL can declare aggregate eligibility universes. Neither certifies an election. This paper records the ownership contract, the two hard invariants, and the deferral decision: RSTAT must not change verifier pass/fail results, and RROLL must not publish private voter-level rolls.

1 Introduction

Statistical and forensic analysis is valuable public evidence. It can find patterns in counts, histories, plans, or spatial context that deserve explanation. Aggregate eligibility universes are also valuable. They can say which registration, eligibility, or ballot-style populations a count record was compared against.

Those two kinds of evidence are not count certification. If analysis is treated as proof, a model score can be laundered into an official verdict without the ledger, audit, or certification record doing that work. If eligibility evidence is published at the wrong level, a transparency artifact can become a private voter-roll disclosure. The civic-evidence boundary therefore reserves two layers and keeps both narrow.

RSTAT is the reserved layer for statistical and forensic analysis outputs. RROLL is the reserved layer for aggregate registration, eligibility, and ballot-style universes. This paper does not describe implemented crates, functions, fixtures, or command-line surfaces for either layer. It records the ownership definitions, the deferral decision, and the rule that governs later implementation: RSTAT can identify patterns that need explanation. RROLL can declare aggregate eligibility universes. Neither certifies an election.

2 Ownership Contract

The downstream boundary specification assigns RSTAT and RROLL separate owners because their evidence has different risks [Della-Libera, 2026a]. RSTAT owns analysis outputs over other packages. RROLL owns aggregate eligibility universes from registration and ballot-style sources. Neither layer owns the count ledger, audit method, certification action, custody record, or public case bundle.

Layer	May assert	Must not assert or publish
RSTAT	Model id and version; input package hashes; anomaly scores; uncertainty intervals; diagnostics; caveats; reproducible model transcripts; patterns that need explanation.	Certification pass/fail; changed verifier outcomes; official findings by implication; privacy-sensitive small-cell outputs without a disclosure policy. Hard invariant: RSTAT cannot change verifier pass/fail results.
RROLL	Aggregate registration snapshots; aggregate eligibility universes; ballot-style universe records; source hashes; privacy caveats; derived aggregate eligibility summaries.	Private voter-level rolls; voter-roll custody or access-control claims; count certification; individual eligibility determinations. Hard invariant: RROLL cannot publish private voter-level rolls.

The access-pattern specification gives the intended read direction [Della-Libera, 2026a]. RSTAT reads normalized RCOUNT totals and CVR summaries, RHIST unit lineage for time series, and RPLAN or RCTX records for spatial and district features. RROLL reads registration, eligibility, and ballot-style sources, with optional RCTX unit universes when aggregate records need geography alignment. These reads are references to evidence owned elsewhere; they do not transfer ownership of those facts into RSTAT or RROLL.

3 Deferral and Boundary

Both layers are deferred. The downstream evidence specification explicitly says not to implement RSTAT, RROLL, or the other downstream packages until their access patterns and claim boundaries keep earlier packages from absorbing their responsibilities [Della-Libera, 2026a]. That deferral is part of the design, not a missing implementation note.

RSTAT should be built only after enough normalized RCOUNT and RHIST inputs exist to make statistical analysis reproducible rather than anecdotal. A forensic report over unstable or one-off inputs would encourage readers to treat a model output as stronger than the evidence beneath it. RSTAT therefore waits for stable package hashes, normalized time series, and claim-boundary language that keeps a statistic from becoming certification.

RROLL has a different reason to wait. Aggregate eligibility universes are useful only if the disclosure boundary is known before fixtures exist. The fixture ladder should not teach future implementers to publish rare ballot styles, small cells, or private voter-level rows. RROLL therefore waits for privacy and disclosure thresholds, plus source-hash and caveat rules, before any positive or negative fixture is written.

The operational boundary is simple. RSTAT can identify patterns that need explanation. RROLL can declare aggregate eligibility universes. Neither certifies an election. Certification belongs to the packages and legal actions that bind count verification, audit evidence, and official signoff. Analysis and eligibility universes may inform those conversations, but they do not replace them.

4 Conclusion

RSTAT and RROLL are reservations for future evidence, not claims about current code. RSTAT will own statistical and forensic analysis outputs when the normalized RCOUNT, RHIST, RPLAN, and RCTX inputs are stable enough to support reproducible models. RROLL will own aggregate registration, eligibility, and ballot-style universes when privacy and disclosure thresholds are defined before fixtures.

The two invariants are the durable result of this paper. RSTAT must not change verifier pass/fail results. RROLL must not publish private voter-level rolls. Together they keep useful analysis and useful eligibility context from being confused with count certification, voter-roll custody, or private data release.

References

Gio Della-Libera. Civic evidence layer access patterns and downstream evidence boundaries. Technical report, Apportionment Research, 2026a.

Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026b.